AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Amity University Uttar Pradesh, Greater Noida

Minor Project (ETMN100)

WEEKLY PROGRESS REPORT (WPR) – 6

For the week commencing: 24/08/2026 – 30/08/2026
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Project Title: WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar

Results

4.1 Data Volume and Coverage

Table 1 summarises the data processed by the system as of this reporting period. The live WQMIS crawler retrieved 822 raw public records, yielding 755 scored laboratory reports across Uttar Pradesh (511) and Bihar (311), containing 3,124 discrete parameter readings evaluated against BIS IS 10500. This represents a substantial expansion beyond the initial 112-sample exploratory crawl, though it remains a partial sample rather than exhaustive state coverage. The supplementary historical contamination registry provides broader coverage, contributing 100,690 quality-affected habitation events across 14,044 tracked villages for the 2009–2012 period, and forms the primary basis for the recurrence analysis in Section 4.3.

Metric	Value
Live WQMIS samples retrieved and scored	755 (UP: 511, Bihar: 311)
Historical contamination registry events loaded (2009–2012)	100,690
Villages tracked in geographic hierarchy	14,044
Districts represented in database hierarchy	99
Automated test cases passing	26 / 26
REST API endpoints exposed	9
Recurrence-query optimisation	37 s → 0.58 s (identical output)

4.2 Contamination Severity Findings

Applying the BIS 10500-based scoring engine (Section 3.3) to the retrieved live samples surfaced at least one Critical-band result during this period: a sample from Lucknow, Uttar Pradesh returned an E. coli detection, which under the band-escalation rule is classified as Critical irrespective of its composite numeric score, consistent with the rule's intent of not allowing an acute microbiological hazard to be diluted by unaffected parameters in the same report. Across the full 869 scored samples, 22 fell in the Critical band (2.5%), 323 in the High band (37.2%), and 515 in the Low/Safe band (59.3%). High-band results were dominated by geogenic Fluoride exceedances (≥ 1.5 mg/L) in Unnao and Firozabad, elevated agricultural Nitrates, and dissolved Iron above permissible limits.

4.3 Recurrence and Chronic Hotspot Findings

The recurrence detection module was run against the full historical registry to classify village-parameter combinations into the four verdict categories defined in Section 3.4. Among the results produced, the village of Khokha in Purnia district, Bihar, was flagged as a chronic-hotspot for iron contamination, with 166 recorded exceedance events across the historical registry period — the type of result the recurrence module is specifically designed to surface, since a single-reading severity score alone would not distinguish this village

from one with a single isolated iron exceedance. A full quantitative breakdown of verdict-category counts (chronic-hotspot / recurring / watchlist / no-history) across all 14,044 villages is planned as an extension of this analysis in the final report.

4.4 Comparative Observations (Uttar Pradesh vs. Bihar)

The district-level choropleth view allows visual comparison of sample density and contamination severity by state. With the expanded dataset, state-to-state comparison is supported with greater confidence than in the earlier exploratory phase, though district-level densities still reflect crawler coverage rather than proportional contamination burden. The historical registry analysis in Section 4.3, which draws on materially more complete data, supports comparison with the greatest confidence. Within these caveats, Bihar is skewed toward geogenic heavy metals, with the lower Gangetic plain exhibiting widespread dissolved Iron exceedances and localised Arsenic enrichment; Uttar Pradesh exhibits geogenic Fluoride in central districts combined with agricultural Nitrate pollution and surface microbial contamination in shallow aquifers.

4.5 System Performance and Validation Results

All twenty-six automated test cases pass, covering scoring correctness, API contract behaviour, WQMIS parser edge cases, and recurrence logic including a PostgreSQL integration suite. The nine REST endpoints and the three-view dashboard (priority queue, state comparison, district map) were verified to function correctly against both the live and offline-snapshot data paths. The recurrence-ranking performance issue identified during development (Section 3.7) was resolved, reducing execution time from approximately 37 seconds to 0.58 seconds across the full 14,044-village set, with output verified identical to the original implementation via regression tests.